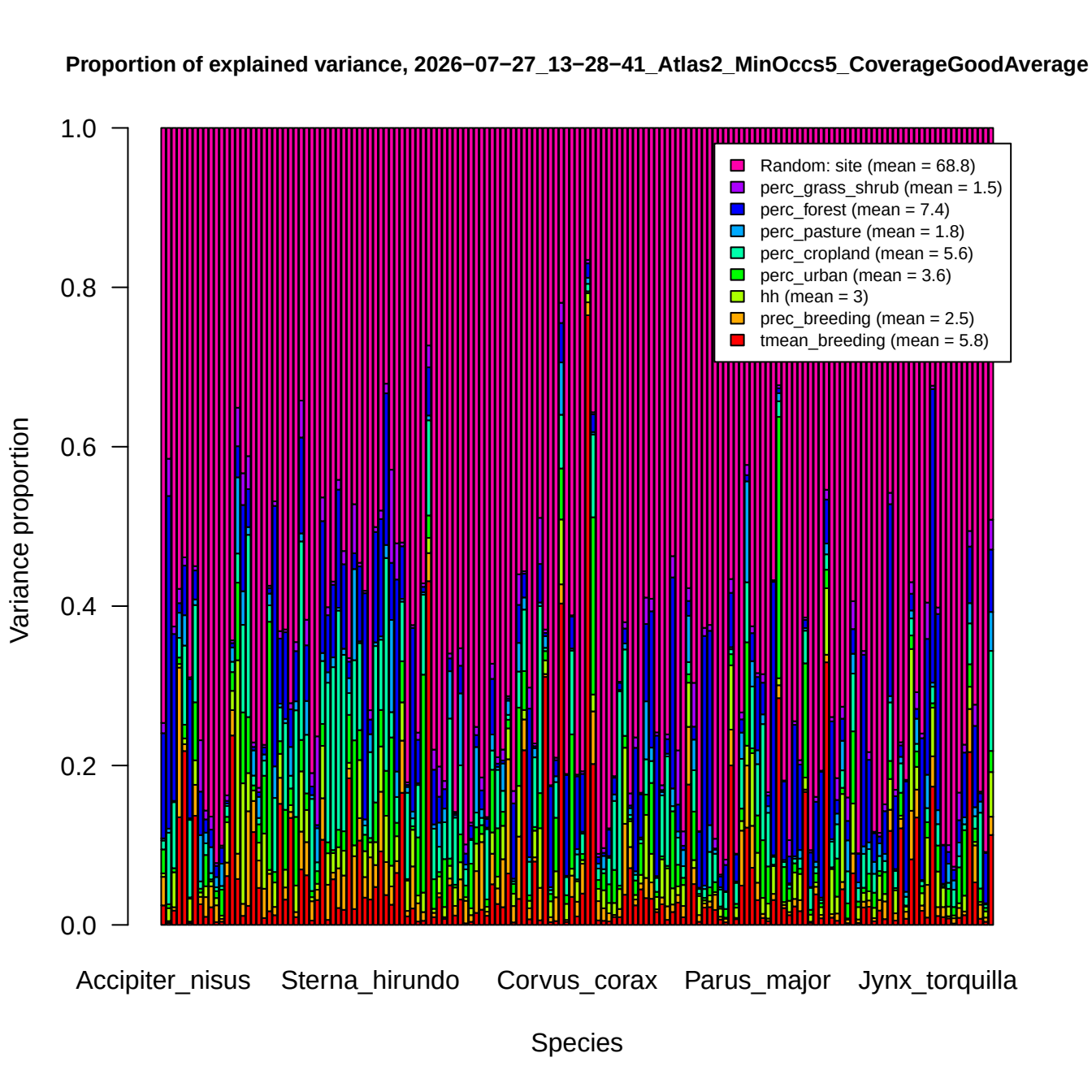
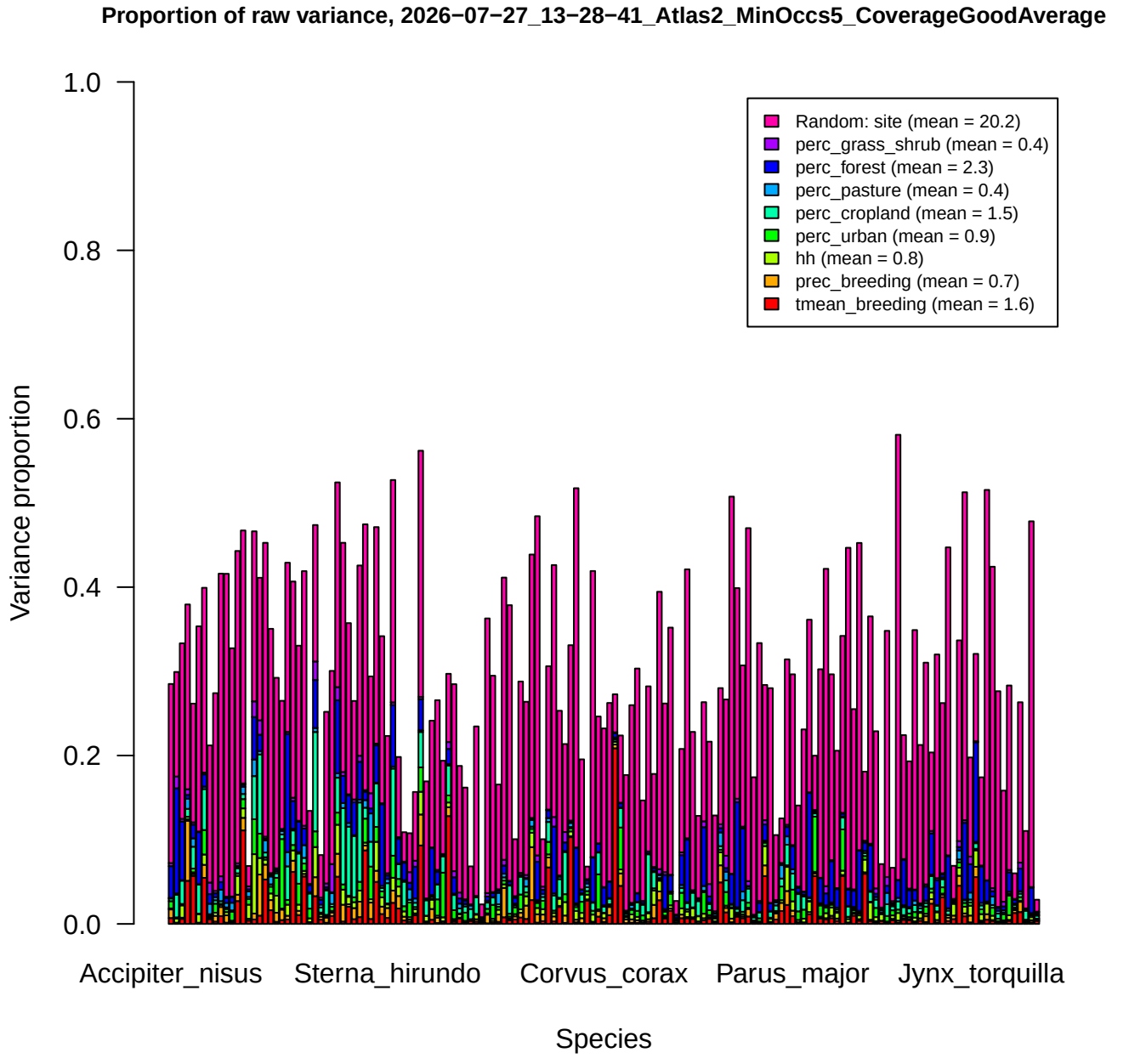
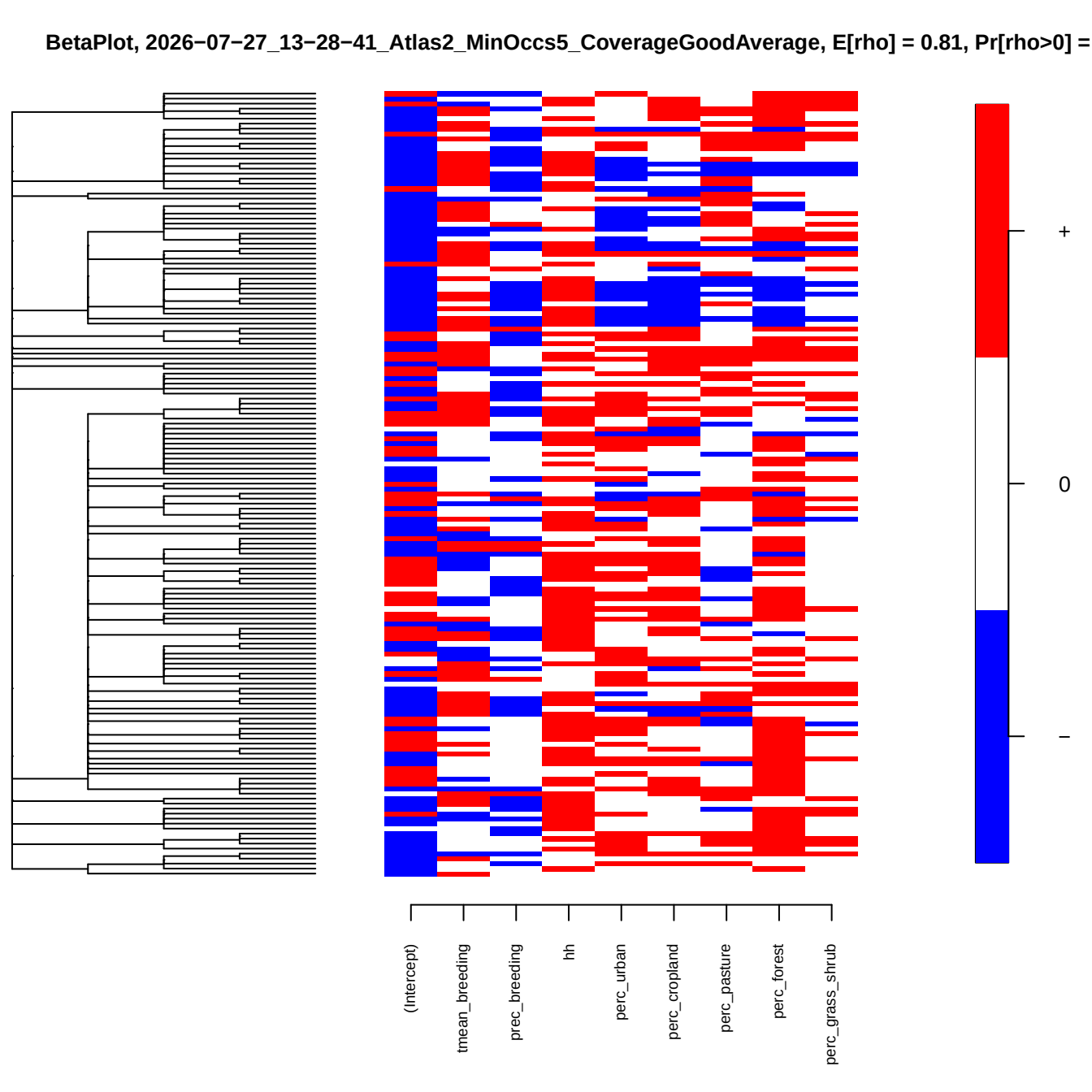
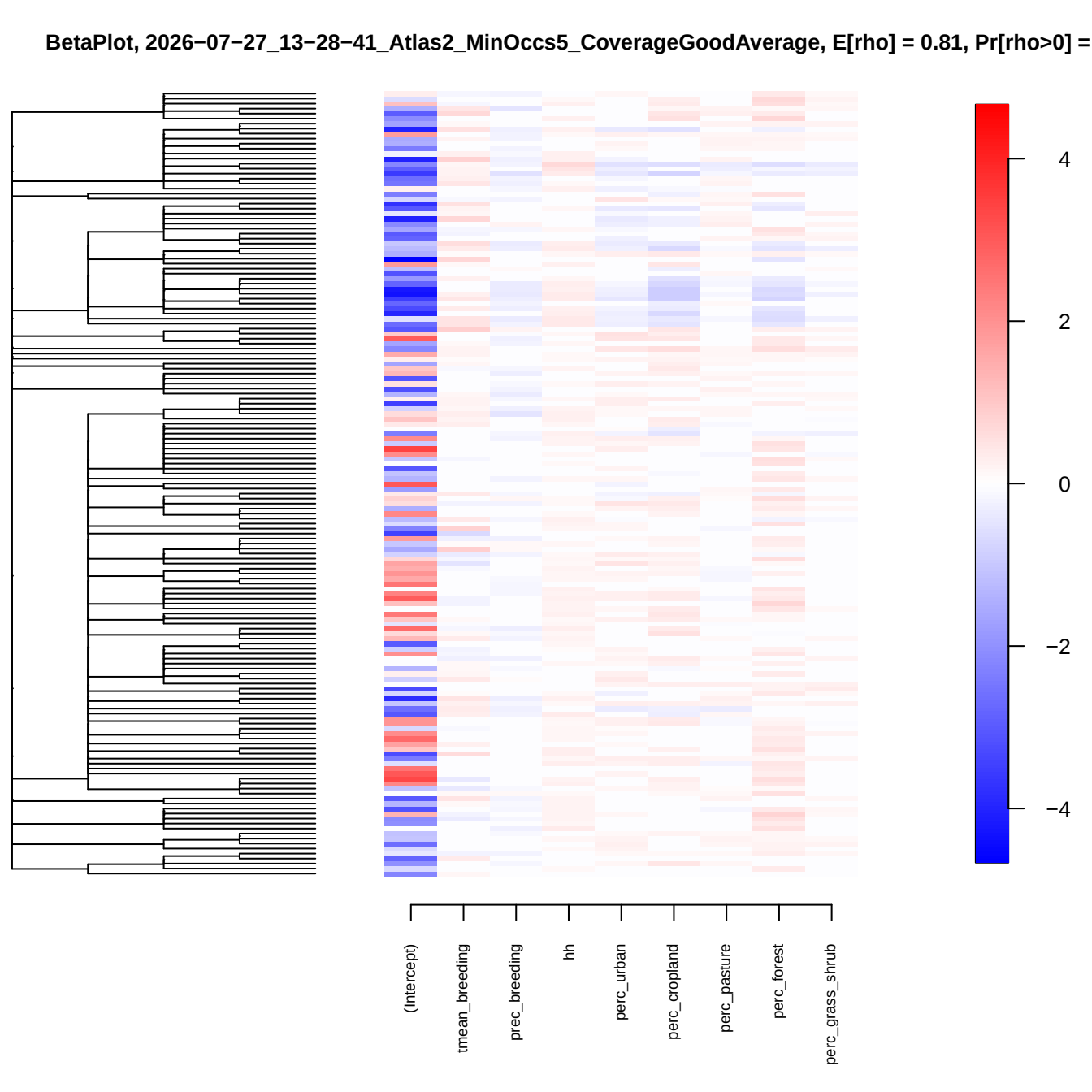
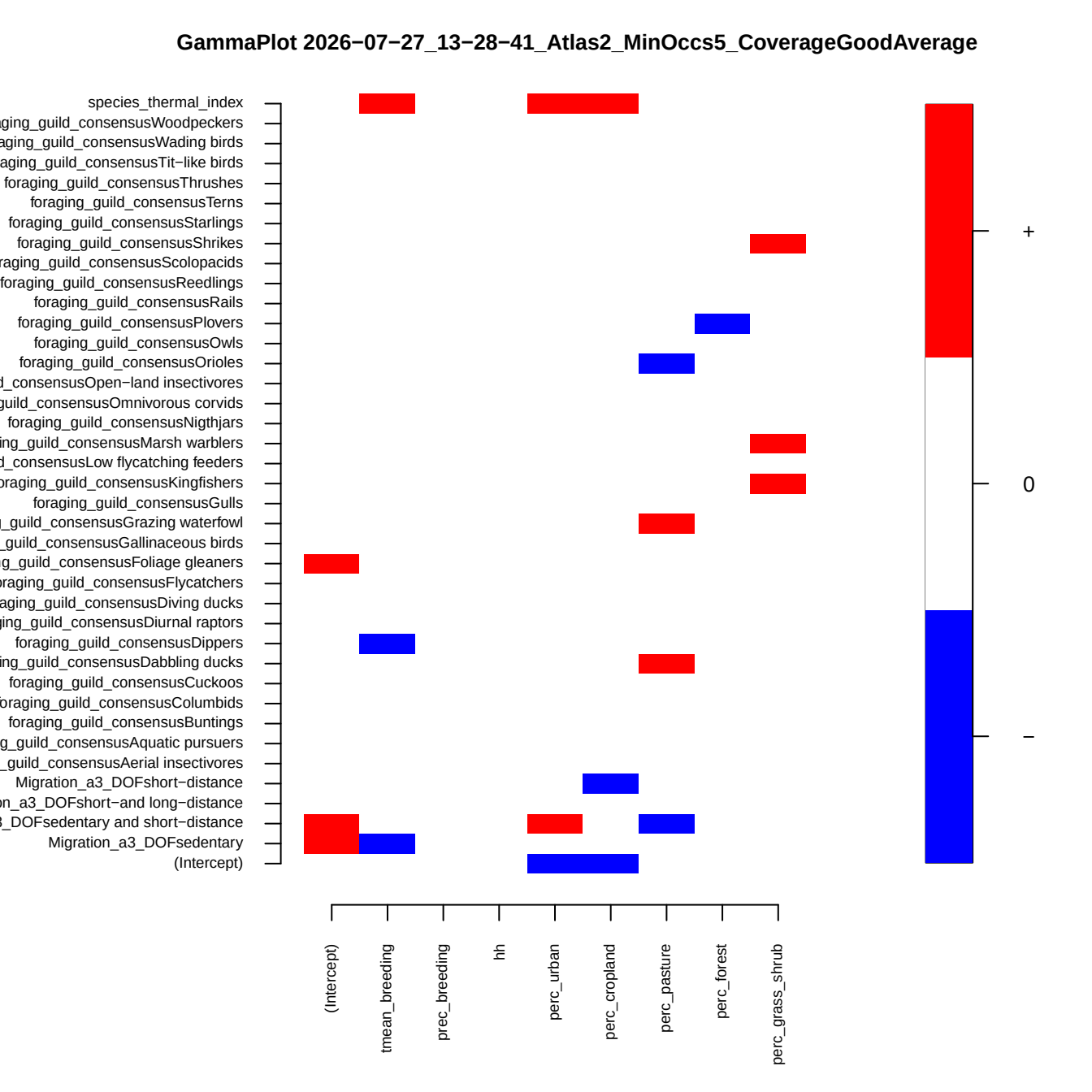
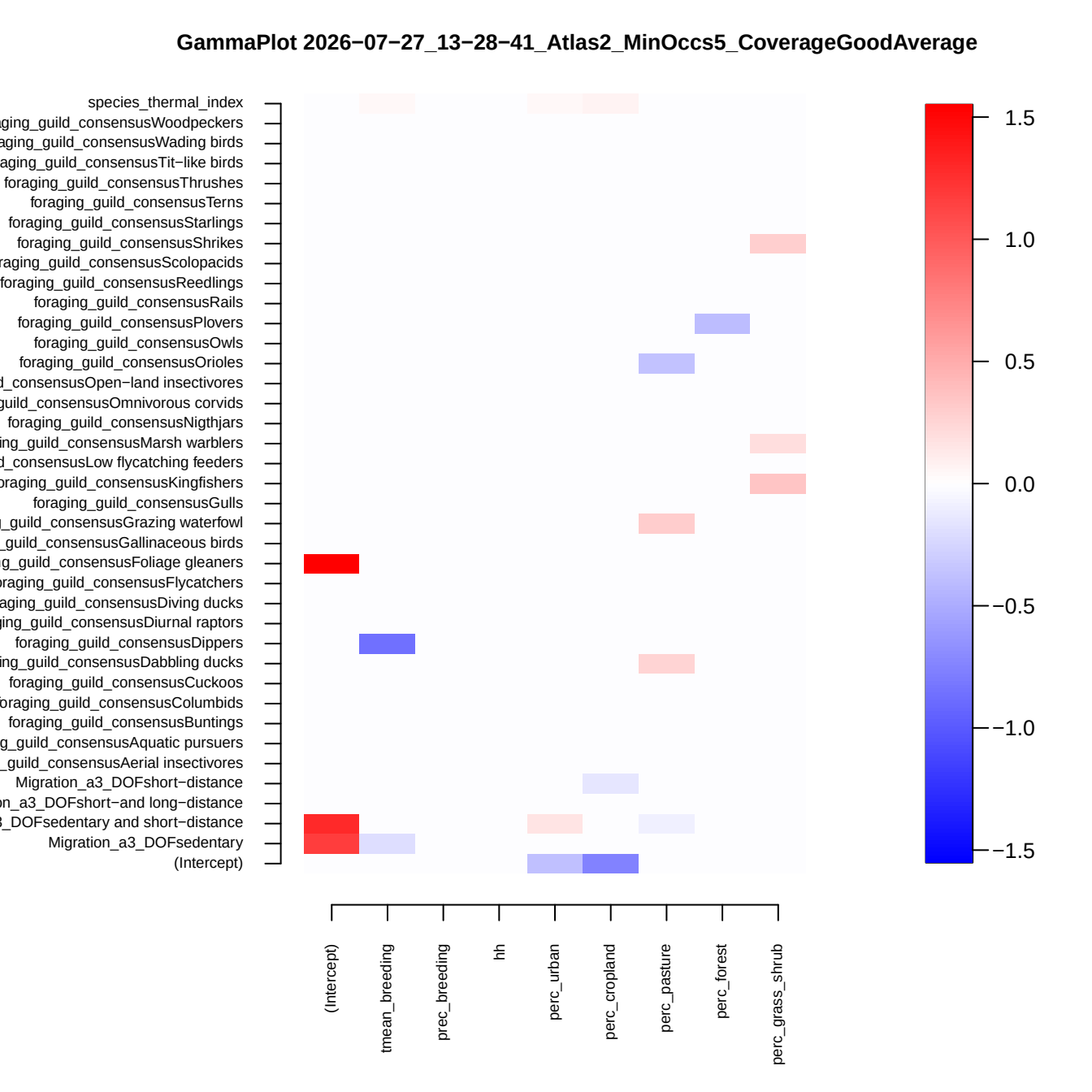

Proportion of explained variance, 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage
Variance proportion
1.0
0.8
0.6
0.4
0.2
0.0
Random: site (mean = 68.8)
perc_grass_shrub (mean = 1.5)
perc_forest (mean = 7.4)
perc_pasture (mean = 1.8)
perc_cropland (mean = 5.6)
perc_urban (mean = 3.6)
hh (mean = 3)
prec_breeding (mean = 2.5)
tmean_breeding (mean = 5.8)
Accipiter_nisus
Sterna_hirundo
Corvus_corax
Parus_major
Jynx_torquilla
Species

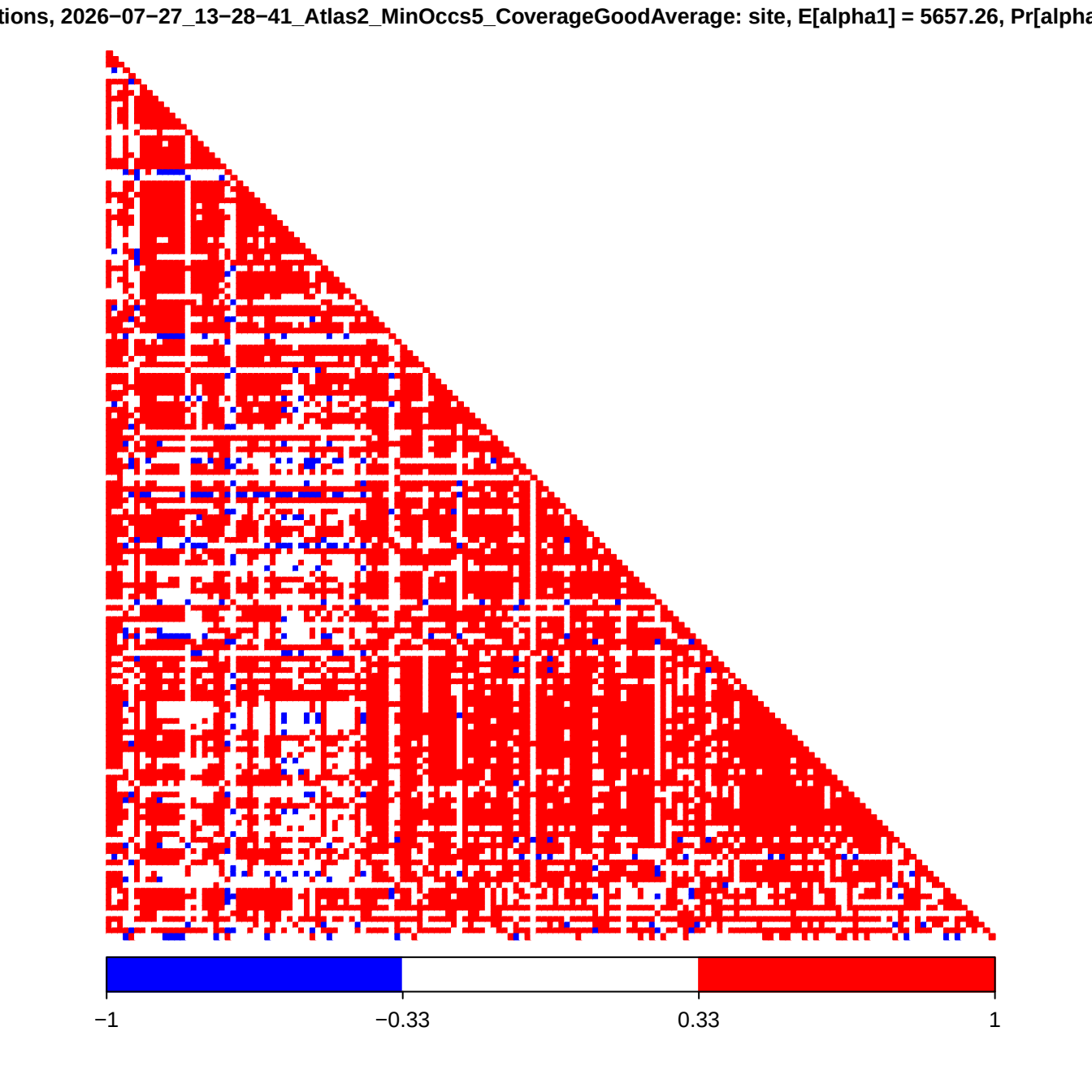

Proportion of raw variance, 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage
Random: site (mean = 20.2)
perc_grass_shrub (mean = 0.4)
perc_forest (mean = 2.3)
perc_pasture (mean = 0.4)
perc_cropland (mean = 1.5)
perc_urban (mean = 0.9)
hh (mean = 0.8)
prec_breeding (mean = 0.7)
tmean_breeding (mean = 1.6)
Variance proportion
1.0
0.8
0.6
0.4
0.2
0.0
Accipiter_nisus
Sterna_hirundo
Corvus_corax
Parus_major
Jynx_torquilla
Species


BetaPlot, 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage, E[rho] = 0.81, Pr[rho>0] =
+
0
−
(Intercept)
tmean_breeding
prec_breeding
hh
perc_urban
perc_cropland
perc_pasture
perc_forest
perc_grass_shrub


BetaPlot, 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage, E[rho] = 0.81, Pr[rho>0] =
4
2
0
−2
−4
(Intercept)
tmean_breeding
prec_breeding
hh
perc_urban
perc_cropland
perc_pasture
perc_forest
perc_grass_shrub


GammaPlot 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage
species_thermal_index
aging_guild_consensusWoodpeckers
aging_guild_consensusWading birds
aging_guild_consensusTit−like birds
foraging_guild_consensusThrushes
foraging_guild_consensusTerns
foraging_guild_consensusStarlings
foraging_guild_consensusShrikes
raging_guild_consensusScolopacids
foraging_guild_consensusReedlings
foraging_guild_consensusRails
foraging_guild_consensusPlovers
foraging_guild_consensusOwls
foraging_guild_consensusOrioles
d_consensusOpen−land insectivores
guild_consensusOmnivorous corvids
foraging_guild_consensusNigthjars
ing_guild_consensusMarsh warblers
d_consensusLow flycatching feeders
oraging_guild_consensusKingfishers
foraging_guild_consensusGulls
_guild_consensusGrazing waterfowl
_guild_consensusGallinaceous birds
g_guild_consensusFoliage gleaners
oraging_guild_consensusFlycatchers
aging_guild_consensusDiving ducks
ging_guild_consensusDiurnal raptors
foraging_guild_consensusDippers
ing_guild_consensusDabbling ducks
foraging_guild_consensusCuckoos
oraging_guild_consensusColumbids
foraging_guild_consensusBuntings
g_guild_consensusAquatic pursuers
_guild_consensusAerial insectivores
Migration_a3_DOFshort−distance
n_a3_DOFshort−and long−distance
3_DOFsedentary and short−distance
Migration_a3_DOFsedentary
(Intercept)
+
0
−
(Intercept)
tmean_breeding
prec_breeding
hh
perc_urban
perc_cropland
perc_pasture
perc_forest
perc_grass_shrub


GammaPlot 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage
species_thermal_index
aging_guild_consensusWoodpeckers
aging_guild_consensusWading birds
aging_guild_consensusTit−like birds
foraging_guild_consensusThrushes
foraging_guild_consensusTerns
foraging_guild_consensusStarlings
foraging_guild_consensusShrikes
raging_guild_consensusScolopacids
foraging_guild_consensusReedlings
foraging_guild_consensusRails
foraging_guild_consensusPlovers
foraging_guild_consensusOwls
foraging_guild_consensusOrioles
d_consensusOpen−land insectivores
guild_consensusOmnivorous corvids
foraging_guild_consensusNigthjars
ing_guild_consensusMarsh warblers
d_consensusLow flycatching feeders
oraging_guild_consensusKingfishers
foraging_guild_consensusGulls
_guild_consensusGrazing waterfowl
_guild_consensusGallinaceous birds
g_guild_consensusFoliage gleaners
oraging_guild_consensusFlycatchers
aging_guild_consensusDiving ducks
ging_guild_consensusDiurnal raptors
foraging_guild_consensusDippers
ing_guild_consensusDabbling ducks
foraging_guild_consensusCuckoos
oraging_guild_consensusColumbids
foraging_guild_consensusBuntings
g_guild_consensusAquatic pursuers
_guild_consensusAerial insectivores
Migration_a3_DOFshort−distance
n_a3_DOFshort−and long−distance
3_DOFsedentary and short−distance
Migration_a3_DOFsedentary
(Intercept)
(Intercept)
tmean_breeding
prec_breeding
hh
perc_urban
perc_cropland
perc_pasture
perc_forest
perc_grass_shrub
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5


tions, 2026−07−27_13−28−41_Atlas2_MinOccs5_CoverageGoodAverage: site, E[alpha1] = 5657.26, Pr[alpha
−1
−0.33
0.33
1